The General Theory of the Cooling Dividend Function

A Theoretico-Analytical Framework for the Marginal Economic Value of Cooling

Lord Soumadeep Ghosh with ChatGPT (OpenAI)

Kolkata, India

Abstract

Cooling is conventionally examined through engineering efficiency, energy consumption, infrastructure design, and environmental consequences. Its economic contribution, however, requires a more general analytical treatment. Cooling can relax thermal constraints on production, preserve productive assets, improve the conditions under which labour and human capital operate, support infrastructure reliability, and modify the feasible allocation of economic resources.

This paper develops the Cooling Dividend Function as a general theoretico-analytical framework for representing the incremental economic value attributable to cooling services. It distinguishes the Gross Cooling Dividend, the Marginal Cooling Dividend, and the Net Cooling Dividend, and derives their relationships under explicitly stated assumptions. The analysis examines nonlinear returns, threshold effects, saturation, thermal exposure, cooling dividend elasticity, sectoral aggregation, complementarity, substitutability, dynamic accumulation, uncertainty, and optimal cooling allocation.

The paper also introduces the Cooling Dividend Frontier, representing the maximum attainable economic dividend from cooling under specified resource, technological, thermal, and architectural constraints. It establishes conditional propositions concerning marginal cooling value and optimal allocation while distinguishing mathematical results from empirical hypotheses.

The central proposition is that cooling should not be maximized independently of its costs and alternative uses of resources. Rather, cooling should be allocated until its marginal social economic value is equal to its marginal social opportunity cost, subject to technological, institutional, environmental, and distributional constraints.

The treatise ends with "The End"

Keywords: Economic Cooling, Cooling Dividend Function, Marginal Cooling Dividend, Net Cooling Dividend, Thermal Exposure, Cooling Architecture, Resource Allocation, Economic Productivity, Social Welfare, Cooling Dividend Frontier.

Contents

1 Introduction and Motivation **1**

2 Conceptual Foundations **2**
2.1 Cooling as an Economic Service 2
2.2 The Economic Counterfactual 3
2.3 The Nature of Economic Value 3

3 The General Cooling Dividend Function **4**
3.1 Definition 4
3.2 Interpretation 4
3.3 The Zero-Cooling Reference Point 4
3.4 The Cooling Dividend as a Conditional Function 4

4 The Marginal Cooling Dividend **5**
4.1 Definition . . . 5
4.2 Economic Interpretation . . . 5
4.3 The Second Derivative . . . 6
4.4 A Conditional Proposition on Marginal Returns . . . 6

5 Nonlinearities, Thresholds, and Saturation **7**
5.1 The Problem of Linear Approximation . . . 7
5.2 Threshold Effects . . . 7
5.3 Saturation Effects . . . 7
5.4 A Logistic Illustration . . . 7
5.5 Multiple Thermal Thresholds . . . 8

6 Thermal Exposure and the Cooling Dividend **8**
6.1 Thermal Exposure . . . 8
6.2 Thermal Dependence . . . 8
6.3 Proposition on Thermal Exposure . . . 9
6.4 Thermal Exposure as an Economic State Variable . . . 9

7 The Cooling Dividend Elasticity **10**
7.1 Definition . . . 10
7.2 Interpretation . . . 10
7.3 Elasticity and Functional Form . . . 10

8 Sectoral Cooling Dividend Functions **10**
8.1 Heterogeneous Economic Sectors . . . 10
8.2 Sectoral Marginal Dividends . . . 11
8.3 Sectoral Heterogeneity . . . 11

9 Complementarity and Substitutability **12**
9.1 Cooling Complementarity . . . 12
9.2 Cooling Substitutability . . . 12
9.3 Network Effects . . . 12

10 The Aggregate Cooling Dividend **13**
10.1 General Aggregation . . . 13
10.2 The Interaction Term . . . 13
10.3 Aggregation and Double-Counting . . . 13
10.4 The Aggregate Marginal Cooling Dividend . . . 14

11 The Net Cooling Dividend **14**
11.1 The Need to Account for Opportunity Costs . . . 14
11.2 Definition . . . 14
11.3 Marginal Net Cooling Dividend . . . 15
11.4 Interpretation . . . 15

12 Optimal Cooling and Social Welfare **15**
12.1 The Single-Variable Optimization Problem . . . 15
12.2 Second-Order Condition . . . 15
12.3 Proposition on Optimal Cooling . . . 16
12.4 Boundary Solutions . . . 16
12.5 Social Welfare Extension . . . 16

13 The Cooling Dividend Frontier **17**
13.1 Definition . . . 17
13.2 Economic Interpretation . . . 17
13.3 Frontier Shifts . . . 17
13.4 Frontier and Opportunity Cost . . . 18

14 Dynamic Cooling Dividend Functions **18**
14.1 The Intertemporal Problem . . . 18
14.2 The Dynamic Net Cooling Dividend . . . 18
14.3 Capital Preservation . . . 18
14.4 Dynamic Marginal Cooling Dividend . . . 19

15 Uncertainty and the Cooling Dividend **19**
15.1 Uncertain Thermal Conditions . . . 19
15.2 Risk and Resilience . . . 19
15.3 Reliability and the Cooling Dividend . . . 20

16 Cooling Architecture and the Dividend Function **20**
16.1 Architecture as a Determinant . . . 20
16.2 Architecture-Dependent Marginal Value . . . 20
16.3 Architectural Efficiency . . . 21

17 General Propositions and Theorems **21**
17.1 Proposition 4. Zero-Cooling Normalization . . . 21
17.2 Proposition 5. Positive Marginal Dividend and Monotonicity . . . 21
17.3 Proposition 6. Marginal Value and Social Cost . . . 22
17.4 Theorem 1. Unique Optimal Cooling under Strict Concavity . . . 22
17.5 Proposition 7. Sectoral Reallocation under a Fixed Cooling Budget . . . 22
17.6 Proposition 8. Optimal Allocation with Sector-Specific Costs . . . 23
17.7 Theorem 2. The Cooling Dividend Frontier and Resource Expansion . . . 23

18 Theoretical Implications **24**
18.1 Cooling as a Source of Economic Productivity . . . 24
18.2 Cooling as an Allocation Problem . . . 24
18.3 Cooling as an Architectural Problem . . . 24
18.4 Cooling and Economic Resilience . . . 24
18.5 Cooling and Economic Expansion . . . 25

19 Empirical Research Agenda **25**
19.1 Firm-Level Analysis . . . 25
19.2 Household-Level Analysis . . . 26
19.3 Urban and Regional Analysis . . . 26
19.4 Cross-Country Analysis . . . 26
19.5 Identification Challenges . . . 27
19.6 Measurement of Effective Cooling . . . 27

20 Limitations and Scope Conditions **27**
20.1 Counterfactual Dependence . . . 28
20.2 Measurement Dependence . . . 28
20.3 System Boundaries . . . 28
20.4 General-Equilibrium Effects . . . 28
20.5 Nonlinear and Non-Convex Relationships . . . 28
20.6 Distributional Effects . . . 28
20.7 Environmental Constraints . . . 28

21 Conclusion **28**

1 Introduction and Motivation

Cooling is an economic service. It consumes resources, requires infrastructure, depends upon technological arrangements, and produces benefits that may extend across households, firms, markets, and entire economies.

The economic significance of cooling cannot therefore be represented exclusively by the quantity of electricity consumed, the installed capacity of cooling equipment, or the engineering efficiency of an individual device.

The relevant economic question is broader:

> What is the incremental economic value generated by an additional quantity of effective cooling services, under specified thermal, technological, and resource-allocation conditions?

This paper addresses that question through the Cooling Dividend Function.

The term *dividend* denotes the incremental economic value attributable to cooling relative to an explicitly defined counterfactual. It does not necessarily denote a monetary payment, financial distribution, or realized accounting profit. It represents an economic benefit that may take the form of additional output, preserved productive capacity, avoided losses, improved household welfare, or other consistently measured economic gains.

The analytical framework builds upon the General Theory of Economic Cooling. That framework treats cooling as a cross-sectoral economic service that modifies the thermal constraints under which production, consumption, and infrastructure operate.

The present paper develops a more specific mathematical object within that broader theory.

The distinction is important. A general theory of economic cooling describes the economic role of cooling. The Cooling Dividend Function describes how the economic value attributable to cooling varies with the quantity and quality of cooling services and with the conditions under which those services are delivered.

The paper makes six principal contributions:

1. It formally defines the Gross Cooling Dividend as a counterfactual economic quantity.
2. It defines the Marginal Cooling Dividend as the incremental economic value generated by additional cooling.
3. It distinguishes gross benefits from the Net Cooling Dividend after accounting for social opportunity costs.
4. It develops the relationships between cooling, thermal exposure, nonlinear returns, and economic productivity.
5. It generalizes the function to multiple sectors, allowing for complementarity, substitutability, and network effects.
6. It introduces the Cooling Dividend Frontier and derives conditions for optimal cooling allocation.

The framework is theoretico-analytical. Its mathematical propositions are conditional on their stated assumptions. The paper does not claim that the functional relationships developed here have already been empirically established for every sector, location, or cooling technology.

2 Conceptual Foundations

2.1 Cooling as an Economic Service

Let C denote the effective quantity of cooling services delivered to an economic unit over a specified period.

The term *effective* is essential. Installed cooling capacity is not necessarily equivalent to cooling actually delivered. Cooling services depend upon equipment, energy, water, maintenance, operating conditions, distribution systems, control mechanisms, and institutional arrangements.

A general representation is:

$$C = \Phi(K_C, E_C, W_C, R_C, \mathcal{A}, \mathcal{I}) \tag{1}$$

where:

K_C Installed cooling capital.

E_C Energy allocated to cooling.

W_C Water and other relevant material inputs.

R_C Maintenance and operating resources.

$\mathcal{A}$ Cooling architecture.

$\mathcal{I}$ Institutional and operational arrangements.

The function Φ describes the conversion of cooling inputs and infrastructure into effective cooling services.

It is not assumed that this conversion is linear. Bottlenecks, technological constraints, equipment failures, distribution losses, and resource shortages may cause effective cooling to differ substantially from installed capacity.

2.2 The Economic Counterfactual

An economic dividend must be defined relative to a counterfactual.

Let:

$$Y(C; \mathbf{X}, \Theta, \mathcal{A}) \tag{2}$$

represent the economic outcome under cooling level C, where:

$\mathbf{X}$ Other economic and infrastructural conditions.

Θ Initial thermal exposure.

$\mathcal{A}$ Cooling architecture.

The counterfactual outcome without the relevant cooling intervention is:

$$Y(0; \mathbf{X}, \Theta, \mathcal{A}_0) \tag{3}$$

The subscript on $\mathcal{A}_0$ allows the baseline architecture to differ from the architecture under the cooling intervention.

For a controlled comparison, the relevant non-cooling conditions must be specified and held constant. Where cooling architecture itself changes, that architectural change must be included in the intervention definition.

A counterfactual comparison is meaningful only when the system boundary, measurement unit, time horizon, and treatment of other economic inputs are consistently specified.

2.3 The Nature of Economic Value

The economic outcome Y need not be restricted to gross output.

Depending upon the research question, it may represent:

- Real value added.
- Firm-level productivity.
- Household consumption possibilities.
- Avoided productive asset depreciation.
- Infrastructure reliability.
- Economic surplus.
- A specified social welfare measure.

These measures are not interchangeable.

In particular, an increase in gross output does not automatically establish an equivalent increase in social welfare. Distributional effects, externalities, resource costs, and changes in risk may alter the welfare assessment.

The chosen outcome measure must therefore be explicit.

3 The General Cooling Dividend Function

3.1 Definition

Definition 1. General Cooling Dividend Function.

The Cooling Dividend Function is a mapping that represents the incremental economic value attributable to effective cooling services under specified economic, thermal, technological, and architectural conditions.

It is represented generally by:

$$D_C = \mathcal{D}(C, \Theta, \mathbf{X}, \mathcal{A}) \tag{4}$$

For a controlled counterfactual with a common baseline, the Gross Cooling Dividend is:

$$\boxed{D_C(C) = Y(C; \mathbf{X}, \Theta, \mathcal{A}) - Y(0; \mathbf{X}, \Theta, \mathcal{A})} \tag{5}$$

The function may be extended to include additional variables, including institutional conditions, population characteristics, energy prices, and the distribution of cooling services.

3.2 Interpretation

The Gross Cooling Dividend measures the difference between an economic outcome with cooling and the corresponding counterfactual outcome without the specified cooling intervention.

It is not necessarily equal to:

- The revenue of a cooling equipment manufacturer.
- The expenditure incurred on cooling.
- The engineering efficiency of a cooling device.
- The total value of the economic activity taking place in a cooled environment.
- The full social welfare effect of cooling.

The dividend is an incremental economic quantity.

If cooling preserves output that would otherwise have been lost, the avoided loss forms part of the dividend. If cooling permits additional production, that incremental production may also form part of the dividend. If cooling improves household welfare without increasing measured output, that welfare effect requires an appropriate welfare measure.

3.3 The Zero-Cooling Reference Point

Under the controlled counterfactual definition:

$$D_C(0) = 0 \tag{6}$$

This is a normalization identity, not an empirical proposition.

It follows directly from defining the dividend as the difference between the outcome at C and the outcome at $C = 0$.

The zero-cooling reference point does not imply that an economy without a particular cooling intervention has no natural ventilation, passive cooling, shading, thermal adaptation, or other means of managing heat.

The baseline must specify what counts as the absence of the intervention.

3.4 The Cooling Dividend as a Conditional Function

The same quantity of cooling need not produce the same economic dividend in different environments.

Formally:

$$D_C = \mathcal{D}(C \mid \Theta, \mathbf{X}, \mathcal{A}) \tag{7}$$

The economic dividend may vary with:

- Initial temperature and humidity.
- Duration and intensity of thermal exposure.
- Sectoral production technologies.
- Labour intensity and occupational conditions.
- Infrastructure vulnerability.
- Household income and access.
- Energy availability and prices.
- Cooling architecture and operational reliability.

Consequently, a universal numerical value for the Cooling Dividend cannot be inferred from cooling capacity alone.

4 The Marginal Cooling Dividend

4.1 Definition

Definition 2. Marginal Cooling Dividend.

The Marginal Cooling Dividend is the additional economic value attributable to an infinitesimal increase in effective cooling services, holding the other arguments of the specified function constant.

For a differentiable Cooling Dividend Function:

$$\boxed{MD_C(C) = \frac{\partial D_C(C)}{\partial C}} \tag{8}$$

For a discrete increase in cooling services:

$$MD_C^{\Delta} = \frac{D_C(C + \Delta C) - D_C(C)}{\Delta C} \tag{9}$$

where $\Delta C > 0$.

The derivative is a local measure. The finite difference is an average marginal contribution over the specified interval.

4.2 Economic Interpretation

The Marginal Cooling Dividend answers a different question from the Gross Cooling Dividend.

The Gross Cooling Dividend asks:

> How much additional economic value is attributable to the specified cooling intervention?

The Marginal Cooling Dividend asks:

> How much additional economic value is attributable to one more unit of effective cooling?

A large total dividend does not necessarily imply a large marginal dividend.

For example, an economic unit may already receive substantial benefits from existing cooling infrastructure. An additional unit of cooling may generate relatively little additional value.

Conversely, where a thermal constraint is binding, an additional unit of cooling may produce a disproportionately large benefit.

4.3 The Second Derivative

The second derivative measures the change in the Marginal Cooling Dividend:

$$\frac{\partial^2 D_C}{\partial C^2} = \frac{\partial MD_C}{\partial C} \tag{10}$$

Three broad cases can be distinguished.

Case 1: Diminishing marginal cooling returns.

$$\frac{\partial^2 D_C}{\partial C^2} < 0 \tag{11}$$

The Marginal Cooling Dividend declines as effective cooling increases.

Case 2: Constant marginal cooling returns.

$$\frac{\partial^2 D_C}{\partial C^2} = 0 \tag{12}$$

The Marginal Cooling Dividend remains constant over the relevant interval.

Case 3: Increasing marginal cooling returns.

$$\frac{\partial^2 D_C}{\partial C^2} > 0 \tag{13}$$

The Marginal Cooling Dividend increases as cooling expands.

None of these cases is universally applicable.

The shape of the function depends upon the underlying thermal constraint, production technology, infrastructure, and economic environment.

4.4 A Conditional Proposition on Marginal Returns

Proposition 1. Diminishing Marginal Cooling Dividend.

Suppose that:

1. The Gross Cooling Dividend is twice continuously differentiable over the relevant interval.
2. Additional cooling progressively relaxes a thermal constraint.
3. The economic value of relaxing the remaining constraint declines with the quantity of cooling already delivered.

Then the Marginal Cooling Dividend is decreasing over that interval:

$$\frac{\partial^2 D_C}{\partial C^2} < 0 \tag{14}$$

Proof.

By assumption, the economic value of relaxing the remaining thermal constraint declines as C increases.

The Marginal Cooling Dividend is the derivative of the Gross Cooling Dividend with respect to cooling:

$$MD_C(C) = \frac{\partial D_C}{\partial C} \tag{15}$$

A declining marginal value implies:

$$\frac{\partial MD_C}{\partial C} < 0 \tag{16}$$

Therefore:

$$\frac{\partial^2 D_C}{\partial C^2} < 0 \tag{17}$$

The result follows under the stated assumptions. □

The proposition is conditional. It does not rule out threshold effects, network complementarities, or other situations in which marginal returns initially increase.

5 Nonlinearities, Thresholds, and Saturation

5.1 The Problem of Linear Approximation

A linear Cooling Dividend Function would take the form:

$$D_C(C) = \alpha C \tag{18}$$

where α is a constant.

This representation is mathematically convenient but may be too restrictive for many economic applications.

Thermal constraints can become binding only after a particular exposure level is reached. Cooling may then produce limited benefits below a threshold and substantial benefits after that threshold.

At higher cooling levels, additional services may generate progressively smaller benefits.

A general theory should therefore allow nonlinear functional forms.

5.2 Threshold Effects

Let $\underline{C}$ represent a threshold level of effective cooling.

A stylized threshold function is:

$$D_C(C) = \begin{cases} 0, & C \leq \underline{C} \\ f(C), & C > \underline{C} \end{cases} \tag{19}$$

where $f(C)$ is an increasing function over the relevant interval.

This representation captures situations in which a minimum level of cooling is required before a productive process, infrastructure component, or operating environment can function effectively.

The function may be continuous or discontinuous at the threshold. Its differentiability must be examined rather than assumed.

5.3 Saturation Effects

At sufficiently high cooling levels, the thermal constraint may be largely relaxed.

Additional cooling may then generate diminishing benefits.

A general saturation condition is:

$$\lim_{C \to \infty} \frac{\partial D_C}{\partial C} = 0 \tag{20}$$

This is a possible property of a particular Cooling Dividend Function, not a universal law.

Physical constraints, minimum operating temperatures, process requirements, and other technological conditions may instead impose a finite feasible interval for C.

5.4 A Logistic Illustration

A convenient illustrative function is:

$$D_C(C) = \frac{D_{\max}}{1 + \exp[-a(C - C_0)]} - \frac{D_{\max}}{1 + \exp(aC_0)} \tag{21}$$

where:

$D_{\max}$ A limiting dividend scale.

a A parameter controlling the steepness of the transition.

C_0 The location of the transition.

The subtraction term ensures that the dividend equals zero at $C = 0$.

The function can represent an initially slow response, a rapid increase around a threshold, and eventual saturation.

It is an illustrative functional form. Its appropriateness must be established for the economic application under consideration.

5.5 Multiple Thermal Thresholds

Economic systems may contain several thresholds.

For example, the cooling requirements of a data centre, a manufacturing facility, a hospital, and a residential building may differ substantially.

A general multiple-threshold representation is:

$$D_C(C) = \sum_{j=1}^{m} w_j f_j(C; \underline{C}_j) \tag{22}$$

where:

m Number of relevant thermal or operational thresholds.

w_j Economic weights associated with the respective components.

$\underline{C}_j$ Threshold cooling level for component j.

This formulation permits heterogeneous cooling requirements within a single economic system.

6 Thermal Exposure and the Cooling Dividend

6.1 Thermal Exposure

Let Θ represent adverse thermal exposure.

For a simple analytical treatment, Θ may be represented as a scalar. In more realistic applications, it should be treated as a vector:

$$\boldsymbol{\Theta} = (\Theta_1, \Theta_2, \ldots, \Theta_n) \tag{23}$$

Its components may represent:

- Ambient temperature.
- Relative humidity.
- Wet-bulb temperature.
- Duration of exposure.
- Temperature variability.
- Heat-wave intensity.
- Night-time thermal conditions.
- Indoor thermal exposure.

The appropriate variables depend upon the economic activity being examined.

6.2 Thermal Dependence

The Cooling Dividend Function can be written as:

$$D_C = \mathcal{D}(C, \Theta) \tag{24}$$

The marginal cooling dividend is:

$$MD_C = \frac{\partial \mathcal{D}}{\partial C} \tag{25}$$

The dependence of marginal cooling value upon thermal exposure is captured by:

$$\frac{\partial^2 \mathcal{D}}{\partial C \partial \Theta} \tag{26}$$

If:

$$\frac{\partial^2 \mathcal{D}}{\partial C \partial \Theta} > 0 \tag{27}$$

then higher adverse thermal exposure increases the marginal economic value of cooling, holding the other relevant variables constant.

This is a conditional complementarity between thermal exposure and cooling services.

It should not be assumed universally. Extremely high thermal exposure may overwhelm available cooling capacity, disrupt energy supply, or create constraints that cooling alone cannot resolve.

6.3 Proposition on Thermal Exposure

Proposition 2. Thermal Exposure and Marginal Cooling Value.

Suppose that the Cooling Dividend Function is twice continuously differentiable in C and Θ. If:

$$\frac{\partial^2 \mathcal{D}}{\partial C \partial \Theta} > 0 \tag{28}$$

over a specified domain, then:

$$\frac{\partial MD_C}{\partial \Theta} > 0 \tag{29}$$

over that domain.

Proof.

By definition:

$$MD_C = \frac{\partial \mathcal{D}}{\partial C} \tag{30}$$

Differentiating with respect to thermal exposure gives:

$$\frac{\partial MD_C}{\partial \Theta} = \frac{\partial^2 \mathcal{D}}{\partial \Theta \partial C} \tag{31}$$

Under continuous second-order differentiability, the mixed partial derivatives are equal. Therefore:

$$\frac{\partial MD_C}{\partial \Theta} = \frac{\partial^2 \mathcal{D}}{\partial C \partial \Theta} > 0 \tag{32}$$

The result follows. □

6.4 Thermal Exposure as an Economic State Variable

Thermal exposure can also be treated as a state variable that evolves over time.

Let:

$$\Theta_{t+1} = \Psi(\Theta_t, C_t, Z_t) \tag{33}$$

where Z_t represents external climatic and environmental conditions.

The function Ψ may include adaptation, passive cooling, building characteristics, urban form, and other determinants of thermal exposure.

The economic dividend then depends not only upon current cooling but also upon the inherited thermal state.

This distinction is important for long-term infrastructure planning and climate adaptation.

7 The Cooling Dividend Elasticity

7.1 Definition

Definition 3. Cooling Dividend Elasticity.

Where $C > 0$ and $D_C(C) \neq 0$, the Cooling Dividend Elasticity is defined as:

$$\boxed{\varepsilon_{D,C} = \frac{\partial D_C}{\partial C} \frac{C}{D_C}} \tag{34}$$

It measures the local percentage change in the Gross Cooling Dividend associated with a one-percent change in effective cooling services.

Equivalently:

$$\varepsilon_{D,C} = \frac{\partial \ln |D_C|}{\partial \ln C} \tag{35}$$

where the absolute-value logarithm is defined and the dividend is nonzero.

7.2 Interpretation

Three cases can be distinguished when the dividend is positive.

$\varepsilon_{D,C} > 1$ The dividend is locally elastic with respect to cooling.

$\varepsilon_{D,C} = 1$ The dividend is locally unit elastic.

$0 < \varepsilon_{D,C} < 1$ The dividend is locally inelastic with respect to cooling.

A negative elasticity indicates that additional cooling is associated with a reduction in the measured dividend over the relevant interval.

Where $D_C = 0$, the elasticity is undefined. Near zero, it may be numerically unstable. If the dividend is negative, the sign and interpretation of the elasticity require particular care.

7.3 Elasticity and Functional Form

For a power function:

$$D_C(C) = \alpha C^{\beta} \tag{36}$$

where $\alpha > 0$, $C > 0$, and $\beta > 0$, the elasticity is:

$$\varepsilon_{D,C} = \beta \tag{37}$$

The parameter β is constant in this particular functional form.

For a general nonlinear function, elasticity varies with C.

The Cooling Dividend Elasticity therefore provides a dimensionless measure of responsiveness, but it does not independently identify the underlying economic mechanism.

8 Sectoral Cooling Dividend Functions

8.1 Heterogeneous Economic Sectors

Cooling services operate across sectors with different technologies and thermal requirements.

Let i denote an economic sector.

The sectoral Cooling Dividend Function is:

$$D_{C_i} = \mathcal{D}_i(C_i, \Theta_i, \mathbf{X}_i, \mathcal{A}_i) \tag{38}$$

Possible sectors include:

- Manufacturing.

- Agriculture and food storage.
- Healthcare.
- Residential housing.
- Commercial buildings.
- Transport and logistics.
- Electricity infrastructure.
- Data centres and digital infrastructure.
- Public administration and education.

The sectoral functions need not have identical mathematical forms.

A unit of cooling delivered to a hospital may produce a different economic dividend from the same nominal quantity delivered to a warehouse or residential building.

8.2 Sectoral Marginal Dividends

The Marginal Cooling Dividend for sector i is:

$$MD_{C_i} = \frac{\partial D_{C_i}}{\partial C_i} \tag{39}$$

The sectoral marginal dividends provide a basis for comparing alternative allocations, provided that:

1. The units of cooling are comparable or appropriately converted.
2. The economic outcomes are measured consistently.
3. The relevant time horizons are aligned.
4. Cross-sector effects are included or separately accounted for.

A direct comparison of sectoral marginal dividends is not sufficient if important distributional, environmental, or network effects are omitted.

8.3 Sectoral Heterogeneity

Suppose two sectors, i and j, receive the same additional cooling quantity.

Their marginal dividends are:

$$MD_{C_i} = \frac{\partial D_{C_i}}{\partial C_i} \tag{40}$$

and:

$$MD_{C_j} = \frac{\partial D_{C_j}}{\partial C_j} \tag{41}$$

There is no general reason for these derivatives to be equal.

The difference:

$$\Delta MD_{ij} = MD_{C_i} - MD_{C_j} \tag{42}$$

measures the difference in marginal economic value under the specified measurement and allocation conditions.

It does not, by itself, establish which sector should receive priority. Social opportunity costs, equity, resilience, and essential-service obligations may also matter.

9 Complementarity and Substitutability

9.1 Cooling Complementarity

Two cooling services or two sectoral cooling allocations may be complementary.

Let the aggregate dividend be:

$$D_C = \mathcal{D}(C_1, C_2) \tag{43}$$

The cross-partial derivative is:

$$\frac{\partial^2 D_C}{\partial C_1 \partial C_2} \tag{44}$$

If:

$$\frac{\partial^2 D_C}{\partial C_1 \partial C_2} > 0 \tag{45}$$

then the marginal dividend from C_1 increases with C_2, and vice versa, over the specified domain.

This is local complementarity.

For example, cooling in an electricity distribution facility may complement cooling in a manufacturing facility if the reliability of the former enables the latter to operate more effectively.

9.2 Cooling Substitutability

If:

$$\frac{\partial^2 D_C}{\partial C_1 \partial C_2} < 0 \tag{46}$$

then the two cooling services are locally substitutable in their contribution to the aggregate dividend.

For example, alternative cooling technologies may provide overlapping services.

The sign of the cross-partial is a property of the specified economic function. It is not determined solely by the physical similarity of the technologies.

9.3 Network Effects

Cooling can produce indirect benefits through interconnected economic systems.

Consider a simplified chain:

$$C_1 \longrightarrow R_1 \longrightarrow Y_2 \longrightarrow D_C \tag{47}$$

where:

C_1 Cooling in an upstream infrastructure component.

R_1 Reliability of that component.

Y_2 Output of a downstream economic activity.

The aggregate dividend may therefore include network effects.

However, these effects must be measured carefully. Intermediate transactions and downstream output cannot simply be added together without accounting for the possibility of double-counting.

10 The Aggregate Cooling Dividend

10.1 General Aggregation

For n sectors, the aggregate Cooling Dividend may be represented as:

$$D_C^A = \mathcal{D}(C_1, C_2, \ldots, C_n) \tag{48}$$

This general form allows for sectoral interactions.

If sectoral effects are additive and interactions are absent, the aggregate dividend reduces to:

$$\boxed{D_C^A = \sum_{i=1}^{n} D_{C_i}} \tag{49}$$

The additive expression is valid only when the component dividends are measured on a consistent basis and do not double-count common economic effects.

10.2 The Interaction Term

A second-order representation of the aggregate function around a reference allocation is:

$$\begin{aligned} D_C^A \approx D_0 + \sum_{i=1}^{n} D_i C_i + \frac{1}{2} \sum_{i=1}^{n} D_{ii} C_i^2 \\ + \sum_{i<j} D_{ij} C_i C_j \end{aligned} \tag{50}$$

where the coefficients represent derivatives evaluated at the reference point.

The interaction term:

$$\sum_{i<j} D_{ij} C_i C_j \tag{51}$$

captures local cross-sector relationships.

Positive interaction coefficients represent local complementarity, while negative coefficients represent local substitutability.

The approximation is local and should not be treated as a globally valid representation without further assumptions.

10.3 Aggregation and Double-Counting

The aggregate dividend must be defined at a specified level of economic accounting.

Suppose cooling in an electricity facility prevents a power interruption, which allows a manufacturing firm to maintain production.

The resulting benefits may be reflected in:

- Avoided electricity-system losses.
- Avoided manufacturing downtime.
- Preserved supply-chain activity.
- Household and consumer benefits.

These benefits are related, and some may overlap.

An aggregate calculation must distinguish final value added, intermediate transactions, avoided losses, and welfare effects.

A consistent social accounting framework is therefore necessary.

10.4 The Aggregate Marginal Cooling Dividend

For the general aggregate function:

$$D_C^A = \mathcal{D}(C_1, \ldots, C_n) \tag{52}$$

the marginal dividend associated with sector i is:

$$MD_{C_i}^A = \frac{\partial \mathcal{D}}{\partial C_i} \tag{53}$$

This derivative includes both direct and interaction effects represented in the aggregate function. It may therefore differ from the isolated sectoral marginal dividend.

11 The Net Cooling Dividend

11.1 The Need to Account for Opportunity Costs

The Gross Cooling Dividend measures incremental economic value attributable to cooling. It does not establish that cooling is economically optimal.

Cooling requires resources that have alternative uses.

These may include:

- Capital investment.
- Electricity.
- Water.
- Land and infrastructure.
- Labour and maintenance.
- Materials and equipment.
- Public expenditure and administrative capacity.

The economic value of the next-best alternative use of these resources is part of the opportunity cost of cooling.

11.2 Definition

Definition 4. Net Cooling Dividend.

The Net Cooling Dividend is the Gross Cooling Dividend less the incremental social opportunity cost attributable to the specified cooling intervention.

Let $SC(C)$ represent the incremental social opportunity cost of cooling.

Then:

$$\boxed{N_C(C) = D_C(C) - SC(C)} \tag{54}$$

The cost function should include external costs not already incorporated into the resource opportunity costs.

Care must be taken to avoid double-counting. If a resource price already fully reflects a particular social cost, that same cost should not be subtracted again as a separate externality.

11.3 Marginal Net Cooling Dividend

Differentiating the Net Cooling Dividend gives:

$$\frac{\partial N_C}{\partial C} = \frac{\partial D_C}{\partial C} - \frac{\partial SC}{\partial C} \tag{55}$$

Therefore:

$$\boxed{MNC_C = MD_C - MSC_C} \tag{56}$$

where:

$$MSC_C = \frac{\partial SC}{\partial C} \tag{57}$$

is the Marginal Social Cost of Cooling.

11.4 Interpretation

The Marginal Net Cooling Dividend measures the incremental net economic value generated by an additional unit of cooling.

Three conditions are possible:

$$MD_C > MSC_C \qquad \Rightarrow MNC_C > 0 \tag{58}$$

$$MD_C = MSC_C \qquad \Rightarrow MNC_C = 0 \tag{59}$$

$$MD_C < MSC_C \qquad \Rightarrow MNC_C < 0 \tag{60}$$

These relationships provide the foundation for optimal cooling allocation.

12 Optimal Cooling and Social Welfare

12.1 The Single-Variable Optimization Problem

Suppose the objective is to maximize the Net Cooling Dividend.

The optimization problem is:

$$\boxed{C^* = \arg\max_{C \in \mathcal{C}} \left[D_C(C) - SC(C) \right]} \tag{61}$$

where $\mathcal{C}$ is the feasible set of cooling levels.

If the solution is interior and the objective is differentiable, the first-order condition is:

$$\boxed{D'_C(C^*) = SC'(C^*)} \tag{62}$$

The Marginal Cooling Dividend equals the Marginal Social Cost of Cooling.

This condition does not state that the total Gross Cooling Dividend equals total cooling expenditure. It equates marginal benefit and marginal social cost.

12.2 Second-Order Condition

For a local maximum:

$$D''_C(C^*) - SC''(C^*) < 0 \tag{63}$$

This condition establishes local strict concavity of the objective at the candidate optimum.

A global optimum requires additional conditions, such as concavity of the objective over the feasible domain, or direct comparison of all feasible candidate solutions.

12.3 Proposition on Optimal Cooling

Proposition 3. Marginal Optimality of Cooling.
Suppose:

1. $D_C(C)$ and $SC(C)$ are continuously differentiable.
2. The feasible set $\mathcal{C}$ is a compact interval.
3. The Net Cooling Dividend is strictly concave over $\mathcal{C}$.

Then a unique global maximizer exists.
If that maximizer is interior, it satisfies:

$$D'_C(C^*) = SC'(C^*) \tag{64}$$

Proof.
Since $\mathcal{C}$ is compact and the Net Cooling Dividend is continuous, the extreme value theorem guarantees that a global maximum exists.
Strict concavity guarantees that the global maximizer is unique.
For an interior maximizer of a differentiable function, the first-order necessary condition is:

$$\frac{\partial}{\partial C}\left[D_C(C^*) - SC(C^*)\right] = 0 \tag{65}$$

Therefore:

$$D'_C(C^*) - SC'(C^*) = 0 \tag{66}$$

and hence:

$$D'_C(C^*) = SC'(C^*) \tag{67}$$

The result follows. □

12.4 Boundary Solutions

The optimal cooling level need not be interior.
If the marginal net dividend remains positive throughout the feasible interval, the optimum may occur at the upper boundary.
If it remains negative, the optimum may occur at the lower boundary.
The first-order equality should therefore not be imposed without checking feasibility and boundary conditions.

12.5 Social Welfare Extension

The Net Cooling Dividend may not capture all dimensions of social welfare.
A broader social objective is:

$$\max_{\mathbf{C},\mathbf{X}} \mathcal{W}(\mathbf{C}, \mathbf{X}) \tag{68}$$

subject to:

$$\mathbf{R}_C + \mathbf{R}_X \leq \overline{\mathbf{R}} \tag{69}$$

where:

$\mathbf{C}$ Cooling allocations.

$\mathbf{X}$ Other economic activities.

$\mathbf{R}_C$ Resources allocated to cooling.

$\mathbf{R}_X$ Resources allocated to other activities.

$\overline{\mathbf{R}}$ Total available resources.

The social welfare function may incorporate distribution, access to essential services, health-related welfare, resilience, and environmental consequences.

The welfare formulation is broader than the maximization of measured output.

13 The Cooling Dividend Frontier

13.1 Definition

Definition 5. Cooling Dividend Frontier.

The Cooling Dividend Frontier represents the maximum attainable Gross Cooling Dividend under specified resource, technological, thermal, and architectural constraints.

Let $\mathcal{F}$ denote the feasible set of cooling allocations and architectures.

The frontier is:

$$\boxed{\mathcal{D}^F(\overline{\mathbf{R}}, \Theta) = \max_{(\mathbf{C}, \mathcal{A}) \in \mathcal{F}} \mathcal{D}(\mathbf{C}, \Theta, \mathcal{A})} \tag{70}$$

where $\overline{\mathbf{R}}$ represents the available resource endowment.

The frontier is conditional upon the specified resource endowment, technology, thermal exposure, and feasibility constraints.

13.2 Economic Interpretation

The Cooling Dividend Frontier is not simply a measure of installed cooling capacity.

It represents the maximum economic dividend attainable from cooling under the specified constraints.

Two economies with identical cooling expenditure may occupy different positions relative to their respective frontiers because they differ in:

- Cooling technology.
- Infrastructure quality.
- Distribution efficiency.
- Sectoral allocation.
- Thermal exposure.
- Institutional capacity.
- Operational reliability.

An economy operating below its frontier may be able to increase its Cooling Dividend through improved allocation or architecture without increasing its total cooling resource endowment.

13.3 Frontier Shifts

The frontier may shift outward when improvements in technology, architecture, or institutional capacity allow a higher dividend from the same resource endowment.

Formally, if a change in architecture from $\mathcal{A}_0$ to $\mathcal{A}_1$ satisfies:

$$\mathcal{D}^F(\overline{\mathbf{R}}, \Theta, \mathcal{A}_1) > \mathcal{D}^F(\overline{\mathbf{R}}, \Theta, \mathcal{A}_0) \tag{71}$$

then the attainable Gross Cooling Dividend has increased under the specified conditions.

This is an architectural improvement in economic cooling performance.

13.4 Frontier and Opportunity Cost

The Cooling Dividend Frontier describes attainable gross benefits. It does not, by itself, identify the optimal allocation.

The optimal allocation must account for the opportunity cost of resources and the alternative economic activities that those resources could support.

Consequently, the Cooling Dividend Frontier and the Net Cooling Dividend are complementary analytical objects.

14 Dynamic Cooling Dividend Functions

14.1 The Intertemporal Problem

Cooling infrastructure is often durable capital.

Its benefits and costs may extend over multiple periods.

Let $K_{C,t}$ denote the cooling capital stock at time t.

A standard capital accumulation equation is:

$$K_{C,t+1} = (1 - \delta_C)K_{C,t} + I_{C,t} \tag{72}$$

where:

δ_C Depreciation rate of cooling capital.

$I_{C,t}$ Investment in cooling capital.

Effective cooling services are generated by:

$$C_t = \Phi(K_{C,t}, E_{C,t}, W_{C,t}, R_{C,t}, \mathcal{A}_t) \tag{73}$$

The period-specific Cooling Dividend is:

$$D_{C,t} = \mathcal{D}(C_t, \Theta_t, \mathbf{X}_t, \mathcal{A}_t) \tag{74}$$

14.2 The Dynamic Net Cooling Dividend

The intertemporal objective may be expressed as:

$$\max_{\{I_{C,t}, C_t\}_{t=0}^{T}} \sum_{t=0}^{T} \beta^t \left[D_{C,t} - SC_t \right] \tag{75}$$

where β is a discount factor.

The optimization is subject to the capital accumulation equation, resource constraints, technological feasibility, and other relevant state dynamics.

The objective allows cooling investments to generate benefits over time rather than treating cooling as a purely contemporaneous service.

14.3 Capital Preservation

Cooling may preserve the productive capacity of existing assets.

Suppose productive capital evolves according to:

$$K_{P,t+1} = (1 - \delta_P(C_t, \Theta_t))K_{P,t} + I_{P,t} \tag{76}$$

where the depreciation rate may depend on cooling and thermal exposure.

If:

$$\frac{\partial \delta_P}{\partial C} < 0 \tag{77}$$

then additional cooling reduces the depreciation rate of productive capital over the specified domain.

The resulting economic benefit may arise through avoided capital losses and preserved productive capacity.

Such benefits should be measured over a consistent horizon and should not be added twice if already reflected in the value of future production.

14.4 Dynamic Marginal Cooling Dividend

The dynamic Marginal Cooling Dividend is the change in the present value of the economic outcome associated with an additional unit of cooling at time t.

It can be represented as:

$$MD_{C,t}^{PV} = \frac{\partial}{\partial C_t}\left[\sum_{\tau=t}^{T} \beta^{\tau-t} D_{C,\tau}\right] \tag{78}$$

This expression includes future effects transmitted through state variables, capital preservation, and other dynamic mechanisms.

It is therefore not necessarily equal to the contemporaneous derivative:

$$\frac{\partial D_{C,t}}{\partial C_t} \tag{79}$$

The difference depends upon the future consequences of current cooling decisions.

15 Uncertainty and the Cooling Dividend

15.1 Uncertain Thermal Conditions

Thermal exposure is uncertain.

Let Θ be a random variable or stochastic process.

The expected Gross Cooling Dividend is:

$$\mathbb{E}[D_C(C, \Theta)] \tag{80}$$

If the function is differentiable and regularity conditions permit interchange of differentiation and expectation, then:

$$\frac{\partial}{\partial C}\mathbb{E}[D_C(C, \Theta)] = \mathbb{E}\left[\frac{\partial D_C(C, \Theta)}{\partial C}\right] \tag{81}$$

The expected Marginal Cooling Dividend is therefore the expectation of the state-contingent marginal dividend under these conditions.

15.2 Risk and Resilience

Expected value alone may not capture the importance of cooling in high-impact thermal events.

A cooling system may have relatively modest average benefits but substantial benefits during extreme heat events.

A risk-sensitive objective may take the form:

$$\max_C \{\mathbb{E}[N_C(C, \Theta)] - \lambda \mathcal{R}(N_C)\} \tag{82}$$

where:

$\mathcal{R}$ A specified measure of risk.

λ A parameter representing the weight assigned to that risk.

Possible risk measures include variance, downside deviation, or a tail-risk measure such as conditional value at risk.

The choice of risk measure should reflect the decision problem.

15.3 Reliability and the Cooling Dividend

Installed capacity may not be available when required.

Let p represent the probability that the cooling service is operational when needed.

A simplified expected dividend is:

$$\mathbb{E}[D_C] = pD_C^{\text{available}} + (1-p)D_C^{\text{unavailable}} \tag{83}$$

This representation illustrates why reliability is an economic variable.

A cooling architecture that increases operational availability may increase the expected dividend even if nominal installed capacity remains unchanged.

16 Cooling Architecture and the Dividend Function

16.1 Architecture as a Determinant

Cooling architecture determines how resources are converted into effective cooling services.

It may include:

- Centralized and distributed systems.
- Passive and active cooling.
- Thermal storage.
- Heat recovery.
- Redundancy and backup capacity.
- Control systems and intelligent operation.
- Water and energy integration.
- Spatial distribution and network design.

The Cooling Dividend Function should therefore be written as:

$$D_C = \mathcal{D}(C, \Theta, \mathbf{X}, \mathcal{A}) \tag{84}$$

rather than assuming that cooling quantity alone determines the economic dividend.

16.2 Architecture-Dependent Marginal Value

The Marginal Cooling Dividend is:

$$MD_C = \frac{\partial \mathcal{D}}{\partial C} \tag{85}$$

The influence of architecture on marginal cooling value is:

$$\frac{\partial^2 \mathcal{D}}{\partial C \partial \mathcal{A}} \tag{86}$$

For a scalar architectural index, a positive cross-partial indicates that improved architecture increases the marginal dividend of cooling over the relevant domain.

For a multidimensional architecture, the relationship must be expressed through the appropriate directional derivative or component-wise partial derivatives.

16.3 Architectural Efficiency

Let:

$$\eta_C = \frac{C}{R_C} \tag{87}$$

where R_C represents a specified cooling resource input.

This ratio is a simplified measure of service delivery per unit of resource input. It is not a universal engineering efficiency measure because the units and quality of cooling services must be defined.

An architecture may improve η_C while also changing the quality, reliability, or distribution of cooling.

Consequently, the economic dividend should be assessed through delivered services and economic outcomes rather than through a single efficiency ratio.

17 General Propositions and Theorems

This section consolidates the principal mathematical implications of the framework.

17.1 Proposition 4. Zero-Cooling Normalization

Under the counterfactual definition:

$$D_C(0) = 0 \tag{88}$$

Proof.
By definition:

$$D_C(C) = Y(C) - Y(0) \tag{89}$$

Substituting $C = 0$:

$$D_C(0) = Y(0) - Y(0) = 0 \tag{90}$$

The result follows. □

17.2 Proposition 5. Positive Marginal Dividend and Monotonicity

Suppose that $D_C(C)$ is continuously differentiable over an interval $[0, \overline{C}]$.

If:

$$D'_C(C) > 0 \tag{91}$$

throughout the interval, then the Gross Cooling Dividend is strictly increasing over that interval.

Proof.
For any $C_2 > C_1$, the mean value theorem implies that there exists $c \in (C_1, C_2)$ such that:

$$D_C(C_2) - D_C(C_1) = D'_C(c)(C_2 - C_1) \tag{92}$$

Both terms on the right-hand side are positive.

Therefore:

$$D_C(C_2) > D_C(C_1) \tag{93}$$

The result follows. □

17.3 Proposition 6. Marginal Value and Social Cost

If:

$$MD_C > MSC_C \tag{94}$$

then an infinitesimal increase in cooling raises the Net Cooling Dividend.
If:

$$MD_C < MSC_C \tag{95}$$

then an infinitesimal increase in cooling reduces the Net Cooling Dividend.
Proof.
By definition:

$$N_C(C) = D_C(C) - SC(C) \tag{96}$$

Differentiating:

$$N'_C(C) = MD_C - MSC_C \tag{97}$$

The two conclusions follow directly from the sign of the derivative. □

17.4 Theorem 1. Unique Optimal Cooling under Strict Concavity

Suppose:

1. The feasible cooling set is a nonempty compact interval.
2. The Gross Cooling Dividend and social cost functions are continuous.
3. The Net Cooling Dividend is strictly concave.

Then there exists a unique global optimal cooling level.
If the optimum is interior and the functions are differentiable, it satisfies:

$$MD_C(C^*) = MSC_C(C^*) \tag{98}$$

Proof.
Continuity on a nonempty compact feasible set guarantees existence of a maximizer.
Strict concavity guarantees uniqueness.
At an interior maximizer, the first-order condition for the differentiable objective must hold:

$$N'_C(C^*) = 0 \tag{99}$$

Since:

$$N'_C(C) = MD_C - MSC_C \tag{100}$$

the equality follows. □

17.5 Proposition 7. Sectoral Reallocation under a Fixed Cooling Budget

Consider two sectors with cooling allocations C_1 and C_2, subject to:

$$C_1 + C_2 = \overline{C} \tag{101}$$

Suppose the aggregate dividend is additive and differentiable:

$$D^A_C = D_{C_1}(C_1) + D_{C_2}(C_2) \tag{102}$$

At an interior allocation maximizing the aggregate dividend, the marginal dividends are equal:

$$MD_{C_1} = MD_{C_2} \tag{103}$$

Proof.
The Lagrangian is:

$$\mathcal{L} = D_{C_1}(C_1) + D_{C_2}(C_2) + \lambda(\overline{C} - C_1 - C_2) \tag{104}$$

The first-order conditions are:

$$\frac{\partial D_{C_1}}{\partial C_1} - \lambda = 0 \tag{105}$$

$$\frac{\partial D_{C_2}}{\partial C_2} - \lambda = 0 \tag{106}$$

Therefore:

$$MD_{C_1} = MD_{C_2} = \lambda \tag{107}$$

The result follows. □

This proposition assumes comparable cooling units and excludes additional sector-specific costs and constraints. If those costs differ, the relevant condition equates marginal net dividends rather than gross marginal dividends.

17.6 Proposition 8. Optimal Allocation with Sector-Specific Costs

Suppose two sectors have sector-specific social cost functions $SC_1(C_1)$ and $SC_2(C_2)$, and a fixed total cooling budget.

At an interior allocation maximizing aggregate net dividend:

$$MD_{C_1} - MSC_1 = MD_{C_2} - MSC_2 \tag{108}$$

Proof.
The objective is:

$$N_C^A = D_{C_1}(C_1) - SC_1(C_1) + D_{C_2}(C_2) - SC_2(C_2) \tag{109}$$

Subject to:

$$C_1 + C_2 = \overline{C} \tag{110}$$

The first-order conditions equate each sector's marginal net dividend to the common multiplier associated with the resource constraint.

Hence:

$$MD_{C_1} - MSC_1 = MD_{C_2} - MSC_2 \tag{111}$$

The result follows. □

17.7 Theorem 2. The Cooling Dividend Frontier and Resource Expansion

Suppose the feasible set expands as the available cooling resource endowment increases:

$$\overline{\mathbf{R}}_2 \geq \overline{\mathbf{R}}_1 \tag{112}$$

and all allocations feasible under $\overline{\mathbf{R}}_1$ remain feasible under $\overline{\mathbf{R}}_2$.

Then:

$$\mathcal{D}^F(\overline{\mathbf{R}}_2, \Theta) \geq \mathcal{D}^F(\overline{\mathbf{R}}_1, \Theta) \tag{113}$$

Proof.
The feasible set under the larger resource endowment contains the original feasible set.

Maximizing the same objective over a set that contains the original feasible set cannot produce a lower maximum.

Therefore:

$$\mathcal{D}^F(\overline{\mathbf{R}}_2, \Theta) \geq \mathcal{D}^F(\overline{\mathbf{R}}_1, \Theta) \tag{114}$$

The result follows. □

This theorem concerns the maximum attainable Gross Cooling Dividend. It does not imply that expanding cooling resources necessarily raises the Net Cooling Dividend. Additional resources may have higher opportunity costs.

18 Theoretical Implications

18.1 Cooling as a Source of Economic Productivity

The Cooling Dividend Function provides a formal means of analysing cooling as a determinant of productive capacity.

If thermal constraints reduce effective labour, damage capital, disrupt production, or impair infrastructure reliability, cooling may relax those constraints.

The resulting economic contribution is represented by the difference between the outcome under cooling and the specified counterfactual.

18.2 Cooling as an Allocation Problem

Cooling is not merely a question of how much infrastructure an economy can install.

It is also a question of where cooling services should be allocated.

Where marginal cooling dividends differ across sectors, reallocating a fixed quantity of cooling may increase the aggregate dividend.

However, an allocation that maximizes aggregate measured output may not maximize social welfare.

Essential services, distributional considerations, and resilience objectives may justify allocations that differ from a narrow output-maximizing solution.

18.3 Cooling as an Architectural Problem

The Cooling Dividend Function depends upon the architecture through which cooling services are delivered.

An economy may therefore increase its Cooling Dividend through:

- Improved cooling technologies.
- Better distribution networks.
- Higher reliability.
- Reduced resource losses.
- Better integration with energy infrastructure.
- Improved building design and passive cooling.
- More effective sectoral allocation.

The frontier formulation captures this possibility.

18.4 Cooling and Economic Resilience

Cooling may reduce the sensitivity of economic activity to thermal shocks.

The economic value of cooling may therefore extend beyond average-period output.

It may include avoided disruptions, lower downside risk, and preservation of essential economic functions during extreme thermal conditions.

The relevant valuation horizon and risk measure must be specified.

18.5 Cooling and Economic Expansion

An increase in the Gross Cooling Dividend may contribute to economic expansion.

However, the aggregate effect depends upon the opportunity cost of the resources used, the distribution of benefits, and the wider general-equilibrium response.

Cooling-induced expansion must therefore be distinguished from:

- Redistribution of existing output.
- Substitution between production technologies.
- Displacement of alternative investments.
- Transfers between economic agents.

The Cooling Dividend Function provides a partial-equilibrium starting point. General-equilibrium analysis is required to determine the wider economy-wide effects.

19 Empirical Research Agenda

The framework is theoretico-analytical, but it generates a set of empirical questions.

The following research directions may be used to evaluate its applicability.

19.1 Firm-Level Analysis

Firm-level panel data could be used to investigate whether cooling services are associated with changes in:

- Labour productivity.
- Real value added.
- Production downtime.
- Equipment failure.
- Capital depreciation.
- Operating costs.
- Output volatility.

A generic empirical specification is:

$$Y_{it} = \alpha_i + \lambda_t + \beta C_{it} + \gamma \Theta_{it} + \delta (C_{it} \times \Theta_{it}) + \mathbf{X}'_{it}\boldsymbol{\eta} + \varepsilon_{it} \tag{115}$$

where:

α_i Firm fixed effects.

λ_t Time fixed effects.

C_{it} Effective cooling services.

Θ_{it} Thermal exposure.

$\mathbf{X}_{it}$ Control variables.

ε_{it} Error term.

The interaction coefficient δ provides an empirical test of whether the relationship between cooling and the outcome varies with thermal exposure.

The coefficient should not automatically be interpreted causally. Identification requires an appropriate research design.

19.2 Household-Level Analysis

Household data could be used to investigate:

- Cooling access and household welfare.
- Cooling expenditure and disposable income.
- Thermal exposure and labour supply.
- Cooling access and educational outcomes.
- Energy poverty and cooling deprivation.
- Distributional differences in cooling benefits.

The analysis should distinguish access to cooling equipment from actual cooling services received.

19.3 Urban and Regional Analysis

Urban and regional data could be used to estimate the relationship between thermal exposure, cooling infrastructure, and economic outcomes.

Relevant variables may include:

- Temperature and humidity.
- Wet-bulb temperature.
- Population-weighted thermal exposure.
- Building characteristics.
- Electricity reliability.
- Cooling access.
- Labour-market outcomes.
- Industrial and commercial activity.

City-centre weather observations alone may be inadequate for measuring metropolitan thermal exposure. Spatially distributed observations or gridded reanalysis data should be considered where appropriate.

19.4 Cross-Country Analysis

A cross-country panel could investigate how cooling dividends vary with:

- Climate.
- Income levels.
- Industrial composition.
- Energy-system reliability.
- Urbanization.
- Institutional capacity.
- Cooling technology.
- Infrastructure quality.

A possible specification is:

$$Y_{it} = \alpha_i + \lambda_t + \beta C_{it} + \gamma \Theta_{it} + \delta (C_{it} \times \Theta_{it}) + \mathbf{X}_{it}' \boldsymbol{\eta} + u_{it} \tag{116}$$

The interpretation of the coefficients depends upon the outcome variable, identification strategy, and measurement of cooling services.

19.5 Identification Challenges

A central empirical challenge is that cooling is not randomly allocated.

Firms and households facing greater thermal exposure may invest more in cooling. Wealthier firms and households may also have better access to cooling.

Consequently, observed associations may reflect:

- Reverse causality.
- Omitted-variable bias.
- Selection effects.
- Measurement error.
- Simultaneity.
- Spatial dependence.

Potential research designs include:

- Carefully specified panel fixed-effects models.
- Natural experiments involving cooling infrastructure.
- Difference-in-differences designs where treatment timing and comparison groups are credible.
- Instrumental-variable strategies with defensible exclusion restrictions.
- Regression discontinuity designs where a genuine allocation threshold exists.
- Structural estimation of cooling demand and production relationships.

No identification strategy should be accepted solely because it is technically available. Its assumptions must be defended in the context of the particular application.

19.6 Measurement of Effective Cooling

The measurement of C is fundamental.

Installed cooling capacity should not be treated as an automatic proxy for effective cooling services.

Possible measures include:

- Delivered thermal energy removal.
- Cooling degree-hours delivered.
- Indoor temperature and humidity outcomes.
- Hours within specified thermal comfort or process limits.
- Operational availability.
- Cooling service per unit of floor area or productive capacity.

The appropriate measure will depend upon whether the object of analysis is human thermal comfort, industrial process cooling, infrastructure protection, or another economic service.

20 Limitations and Scope Conditions

The Cooling Dividend Function is a general analytical framework, not a universal empirical production function.

Several limitations must be recognized.

20.1 Counterfactual Dependence

The estimated dividend depends upon the selected counterfactual.

Different baselines may produce different dividend estimates. The counterfactual must therefore be explicitly justified.

20.2 Measurement Dependence

The dividend depends upon the economic outcome being measured.

Real value added, output, household welfare, avoided losses, and social surplus cannot be treated as equivalent measures.

20.3 System Boundaries

A narrow firm-level analysis may omit upstream and downstream effects.

A national analysis may conceal local distributional consequences.

The system boundary must match the research question.

20.4 General-Equilibrium Effects

The partial-equilibrium dividend does not automatically account for changes in prices, wages, energy demand, investment, migration, or other markets.

A general-equilibrium extension may be necessary for economy-wide conclusions.

20.5 Nonlinear and Non-Convex Relationships

Thresholds, discrete investments, network effects, and technological constraints may produce non-convexities.

In such cases, first-order conditions alone may be insufficient to identify a global optimum.

20.6 Distributional Effects

An aggregate positive dividend does not establish that all economic agents benefit.

Distributional analysis is required to identify who receives the benefits and who bears the costs.

20.7 Environmental Constraints

Cooling may generate environmental costs through electricity consumption, refrigerant leakage, water use, material requirements, and waste heat.

These effects should be incorporated where relevant, without double-counting costs already reflected in the opportunity-cost calculation.

21 Conclusion

This paper has developed the Cooling Dividend Function as a theoretico-analytical framework for understanding the incremental economic value attributable to cooling services.

The framework distinguishes three central economic quantities.

First, the Gross Cooling Dividend:

$$D_C(C) = Y(C) - Y(0) \tag{117}$$

This measures the incremental economic outcome attributable to cooling relative to a specified counterfactual.

Second, the Marginal Cooling Dividend:

$$MD_C(C) = \frac{\partial D_C}{\partial C} \tag{118}$$

This measures the additional economic value attributable to one more unit of effective cooling.

Third, the Net Cooling Dividend:

$$N_C(C) = D_C(C) - SC(C) \tag{119}$$

This accounts for the incremental social opportunity cost of cooling.

The analysis has demonstrated that the Cooling Dividend Function may be nonlinear, threshold-dependent, sector-specific, architecture-dependent, and dynamic. Its marginal value may increase or decrease depending upon the economic and thermal conditions under consideration.

The Cooling Dividend Frontier extends the analysis from the value of a given cooling allocation to the maximum attainable dividend under specified resource and technological constraints.

The central optimization condition is:

$$\boxed{MD_C(C^*) = MSC_C(C^*)} \tag{120}$$

subject to feasibility, appropriate boundary conditions, and the assumptions required for optimality.

This condition expresses the principal economic implication of the theory: the economically desirable quantity of cooling is not necessarily the largest physically achievable quantity. It is the quantity whose marginal social economic value is balanced against its marginal social opportunity cost.

The broader implication is that cooling should be examined as an economic allocation problem, a technological problem, an architectural problem, and a problem of resilience.

An economy may obtain a larger Cooling Dividend not merely by installing more cooling capacity, but by improving the conversion of resources into effective cooling services, directing those services toward economically and socially valuable uses, and reducing the costs and risks associated with their provision.

The theory therefore establishes a distinction between the physical expansion of cooling and the economic optimization of cooling.

The final proposition is:

$$\boxed{\text{The objective of economic cooling is not to maximize cooling,}} \tag{121}$$

$$\boxed{\text{but to maximize the sustainable net economic dividend of cooling.}} \tag{122}$$

References

[1] Arrow, K. J., and Debreu, G. (1954). Existence of an Equilibrium for a Competitive Economy. *Econometrica*, 22(3), 265–290.

[2] Debreu, G. (1959). *Theory of Value: An Axiomatic Analysis of Economic Equilibrium*. New York: John Wiley & Sons.

[3] Hotelling, H. (1931). The Economics of Exhaustible Resources. *Journal of Political Economy*, 39(2), 137–175.

[4] Nordhaus, W. D. (1991). To Slow or Not to Slow: The Economics of The Greenhouse Effect. *The Economic Journal*, 101(407), 920–937.

[5] Rosen, S. (1974). Hedonic Prices and Implicit Markets: Product Differentiation in Pure Competition. *Journal of Political Economy*, 82(1), 34–55.

[6] Stern, N. (2007). *The Economics of Climate Change: The Stern Review*. Cambridge: Cambridge University Press.

[7] Tirole, J. (1988). *The Theory of Industrial Organization*. Cambridge, MA: MIT Press.

[8] Varian, H. R. (1992). *Microeconomic Analysis*, 3rd ed. New York: W. W. Norton & Company.

[9] Wooldridge, J. M. (2010). *Econometric Analysis of Cross Section and Panel Data*, 2nd ed. Cambridge, MA: MIT Press.

Glossary

Cooling Service
: The effective cooling delivered to an economic unit through a specified technological and institutional arrangement.

Cooling Dividend Function
: A mapping representing the incremental economic value attributable to cooling under specified thermal, economic, technological, and architectural conditions.

Gross Cooling Dividend
: The incremental economic outcome attributable to cooling relative to a specified counterfactual.

Marginal Cooling Dividend
: The additional economic value attributable to an infinitesimal increase in effective cooling services.

Net Cooling Dividend
: The Gross Cooling Dividend less the incremental social opportunity cost of cooling.

Marginal Social Cost of Cooling
: The additional social opportunity cost associated with one more unit of cooling.

Cooling Dividend Elasticity
: The percentage responsiveness of the Gross Cooling Dividend to a percentage change in effective cooling services.

Cooling Dividend Frontier
: The maximum attainable Gross Cooling Dividend under specified resource, technological, thermal, and architectural constraints.

Thermal Exposure
: The thermal conditions experienced by an economic unit, potentially represented by temperature, humidity, duration, intensity, and other relevant variables.

Cooling Architecture
: The technological, infrastructural, spatial, and institutional arrangement through which cooling resources are converted into effective cooling services.

Threshold Effect
: A nonlinear response in which the economic dividend changes materially after a specified level of cooling or thermal exposure is reached.

Cooling Complementarity
: A relationship in which the marginal dividend of one cooling service increases with the provision of another.

Cooling Substitutability
: A relationship in which the marginal dividend of one cooling service decreases with the provision of another.

Effective Cooling
: Cooling services actually delivered, rather than merely the nominal capacity installed.

The End